MD YUSUF JAWED

B.Tech CSE (Artificial Intelligence & Machine Learning) | Software Development · AI/ML · Full-Stack

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Profile

Final-year B.Tech Computer Science (AI & ML) student who designs, builds, and evaluates applied AI systems, including retrieval-augmented generation (RAG) pipelines with citation-grounded LLM answers, machine learning models, and IoT-enabled healthcare applications. Skilled in Python, FastAPI, and full-stack development, with six months of client-facing web development experience. Seeking an AI/ML or software engineering role.

Education

SRM IST, Kattankulathur – Bachelor of Technology (B.Tech), Computer Science and Engineering (AI & ML) — Expected 2027

Prakritik School, Patna, Bihar – Class XII (CBSE), PCBM — 2022

DPS, Patna, Bihar – Class X (CBSE), PCBM — 2020

Experience

DAPPRUSH – Web Development Intern (Part-time, Remote) | Live Site — Mar 2025 – Aug 2025

Dubai, UAE

- Built and launched the studio's live website in **Framer** on its custom domain: looping-video hero, 7 service video cards, testimonials, FAQ accordion, tickers, contact, and a scrolling gallery.
- Customized the dark cinematic theme and added **scroll-reveal**, hover, and looping animations by adapting code snippets in Framer (triggers, timing, styling, mobile fixes), plus background **audio**.
- Delivered **3+ more portfolio and client websites** in six months, each **responsive**; applied **client feedback** and handled domain setup.

Projects

MedDoc RAG – Citation-Grounded Q&A over Clinical Guidelines | GitHub

Personal project (solo) | Python, FAISS, BM25, FastAPI, Streamlit, Docker, GitHub Actions

Flow: 8 PDFs → Section-Aware Chunks → bge-small + FAISS → Cross-Encoder Rerank → Grounded LLM → Citation Check → FastAPI

- Built a citation-grounded **RAG** system over 8 WHO/CDC/NIH clinical guidelines (1,077 pages) that refuses out-of-scope questions.
- **Hybrid retrieval:** BGE embeddings + **FAISS**, **BM25**, **Reciprocal Rank Fusion**, **cross-encoder reranking**, section-aware chunking.
- **Evaluation:** Hit@k, MRR and **paired bootstrap CIs** on a 48-question golden set; **95.8% Hit@5**, 0.741 MRR.
- **Safety:** relevance gate, citation validation, **prompt-injection** classifier, output validation, **red-teaming**.
- Shipped with **FastAPI**, **Streamlit**, **Docker**, **pytest** and **GitHub Actions CI**; built on pretrained models.

Neuro – EEG-Based Depression & Mood-Disorder Detection using ML and Deep Learning

Academic project | Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, SHAP, PCA, SMOTE, FastAPI

Flow: EEG features → Feature Engineering → StandardScaler → PCA → SMOTE → 9 ML/DL Models → 5-Fold CV + Test → SHAP → FastAPI

- Built an end-to-end **ML pipeline** classifying healthy vs. mood-disorder subjects from **EEG** band-power features (n=361).
- **Preprocessing:** imputation, **StandardScaler**, **PCA**, **SMOTE** (training data only); engineered **FAA** and Theta/Alpha biomarkers.
- Benchmarked 7 **ML** and 2 **DL** models (**ANN, CNN-LSTM**) with **stratified 5-fold CV** and a held-out test set (n=73).
- **XGBoost** and **Voting Ensemble**: 80.82% test accuracy vs. 73.7% majority baseline; **SHAP** explainability, **FastAPI** serving.

HealthMonitor – IoT Patient Monitoring & Alert System | Live Demo

Academic project | ESP32, Embedded C/C++, MAX30102, DS18B20, Firebase, Node.js, Express, Jest, Chart.js

Flow: MAX30102 + DS18B20 → ESP32 → Wi-Fi → Firebase RTDB → Alert Worker (NEWS2 + Baselines) → Express API → Dashboard

- Built an **IoT** system: **ESP32** firmware, **MAX30102** (HR, SpO2), **DS18B20** (temperature), **Firebase Realtime Database**.
- **Real-time alert engine: NEWS2** scoring, adaptive per-patient baselines (z-score), escalation, clinician feedback loop.
- Secured **Node.js/Express REST API** on Vercel: **JWT**, Helmet, rate-limited login, **Firebase Auth**, device API keys.
- Live **Chart.js** dashboard; **40 Jest tests**; academic project, not a medical device.

Healix – AI-Powered 3D Healthcare Dashboard | Live Demo

React, Vite, Tailwind CSS, Three.js, Zustand, OpenRouter, Vercel, Jenkins, Docker, AWS EC2

Flow: React UI → Zustand → React Three Fiber (5 GLB Models) | AI Request → Vercel Serverless Function → OpenRouter

- Built a responsive **React**, **Vite**, **Tailwind CSS** dashboard with **five interactive 3D models** (**Three.js**, **React Three Fiber**).
- **LLM** chat and report summaries via **OpenRouter API** and **Vercel serverless functions** (API key kept server-side).
- State and routing with **Zustand** and **React Router**; lazy-loaded 3D views.
- **Jenkins CI/CD** pipeline on **AWS EC2** with **Docker**, separate from the Vercel deployment.

Additional Projects

- **Silent Stress** – Anonymous academic-stress reporting platform | Next.js, Tailwind CSS, Vercel | Live Demo
- **Expensi** – Bill-splitting and shared-expense tracker | HTML, CSS, JavaScript, Vercel | Live Demo

Skills

Professional Skills: Client Communication, Interpersonal Skills, Remote Collaboration, Presentations, Documentation & Reporting, Written and Spoken English

AI & Machine Learning: Machine Learning, Deep Learning, Feature Engineering, Explainable AI, RAG, LLM Evaluation, Information Retrieval, Scikit-learn, XGBoost, LightGBM, SHAP, PCA, SMOTE

Programming & Data: Python, JavaScript, SQL, Pandas, NumPy, FAISS, Jupyter; DSA, OOP, DBMS, Operating Systems, Computer Networks

Web Development: React, Next.js, Vite, HTML5, CSS3, Tailwind CSS, Three.js, React Three Fiber, Zustand, Framer, Node.js, Express, FastAPI, REST API, JWT

IoT & Hardware: ESP32, Embedded C/C++, Arduino IDE, MAX30102, DS18B20, I2C, 1-Wire

Cloud, Tools & APIs: Vercel, Jenkins (CI/CD), Docker, GitHub Actions, AWS EC2, Firebase, Git

Languages: English (Professional), Hindi (Native), Urdu (Limited Working), German (Elementary), Korean (Elementary)